

Patterns among the Primes - gaps

<https://github.com/fbholt/Primegaps-v2>

$G(p_0^\#) \rightarrow w_g(\lambda)$
Data prep
Cycles of gaps $G(p^\#)$
Population models $w_g(\lambda)$ for all gaps $g \leq 2p_1$

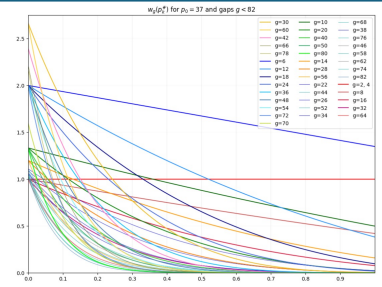
01_Cycles

02_p-rough

11_lambda

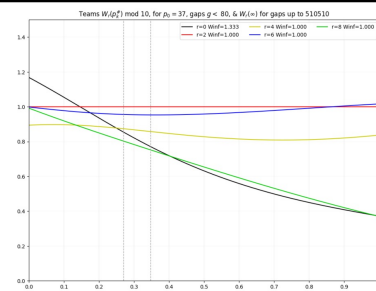
Graphing $w_g(\lambda)$
for all gaps $g \leq 2p_1$
 $\lambda(p_0) = 1$
 $\lambda(p^2) \approx \lambda(p)/2$
 $\lambda(\infty) = 0$

10_wg



Topic: expected biases in last digits
Graphing relative populations $W_r(\lambda)$ by residue r for different bases.

16_bias_lastdigits



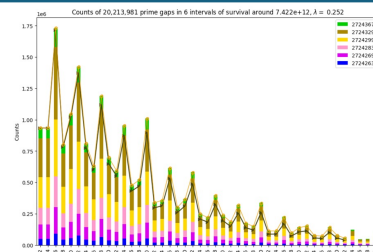
Intervals of survival $\Delta H(p_k)$
Data prep
Prime gaps in intervals of survival
 $\Delta H(p_k) = (p_k^2, p_{k+1}^2]$

03_primeblock

12_DelHprep

Graphing $\Delta H(p_k)$
Histogram of counts of prime gaps.
 $g = 2, 4, 6, 8, 10, 12, \dots$
Bands color-coded by gap
 $g_k = p_{k+1} - p_k$
Compared to estimates based on $w_g(\lambda)$

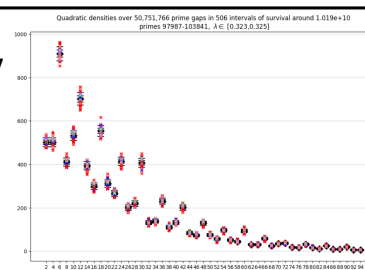
13_DelHdisplay



Topic: quadratic density
Graphing the densities of gaps in intervals of survival.

$$\eta_g(p_k) = n_g(\Delta H(p_k)) / g_k$$

33_quaddensity



Patterns among the Primes

admissible constellations

<https://github.com/fbholt/Primegaps-v2>

$G(p_0^\#) \rightarrow w_s(\lambda)$

Data prep

Population models $w_s(\lambda)$ for select admissible constellations with $|s| \leq 2p_1$

01_Cycles

03_primeblock

Topic:

nonconvex constellations

Studies of Engelsma constellations

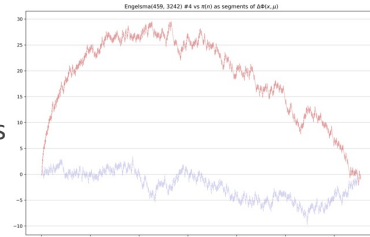
$(J, |s|) = (459, 3242)$

breadth-first searches

[23_nonconvex_prep_Engelsma459](#)

depth-first searches

[24_nonconvex_search_Engelsma459](#)



Graphing $w_s(\lambda)$

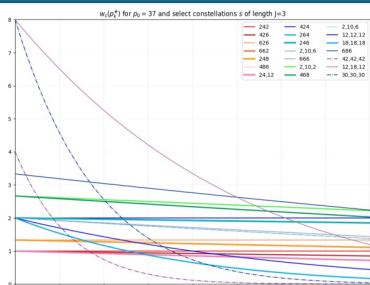
for select constellations of lengths $J=3,5$, with $|s| \leq 2p_1$

$\lambda(p_0) = 1$

$\lambda(p^2) \approx \lambda(p)/2$

$\lambda(\infty) = 0$

[20_ws](#)



Intervals of survival $\Delta H_s(p_k)$

Data prep

Constellations among primes in intervals of survival

$\Delta H(p_k) = (p_k^2, p_{k+1}^2]$

03_primeblock

21_DelHsprep

Graphing $\Delta H(p_k)$

Histogram of counts of constellations among primes.

Bands color-coded by the gap

$g_k = p_{k+1} - p_k$

Compared to estimates based on $w_{s,J}(\lambda)$

[22_DelHsdisplay](#)

